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Portfolio: walt1210.github.io



CAREER PROFILE

Fourth year Bachelor of Science in Computer Science student, majoring in Data Science. Thesis work included building the retrieval-augmented guidance backend on Google Cloud Run and contributing to the model pipeline and Android application of an offline maize disease diagnosis system. Experienced in machine learning, deep learning, retrieval-augmented generation, and agentic artificial intelligence.

SKILLS

- **Programming Languages:** Python, R, Java, C#, PHP, JavaScript, Kotlin, Dart, SQL, HTML, CSS, Assembly
- **Machine Learning and Data Science:** TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, NLTK, HDBSCAN, Matplotlib/Seaborn, data visualization (Power BI and related libraries)
- **Agentic AI and LLM Frameworks:** LangChain (LCEL, prompt chaining, routing), CrewAI (multi-agent orchestration, tool use), Google Gemini API, Pydantic (structured LLM output), RAG pipeline design, prompt engineering
- **Web, Mobile, and Backend Development:** Flask, Android (Kotlin), Flutter, Next.js 15, Supabase
- **Databases:** MySQL, SQLite, Oracle, ChromaDB (vector database)
- **Cloud and DevOps:** Docker, Google Cloud Run, Google Cloud Platform, Google Colab, Linux, Git, Jupyter Notebook, Visual Studio Code
- **Additional Competencies:** Cybersecurity, Networking, Entrepreneurship, AI Prompt Design, Cloud Fundamentals (Microsoft Azure)

PROJECT EXPERIENCE

MAIze: AI-Powered Maize Disease Diagnosis System (Guidance Module) *Thesis Project*

Developed a retrieval-augmented generation system that grounds AI-generated crop disease management advice in a curated agricultural knowledge base. Deployed as a cloud backend supporting a mobile diagnostic application for Filipino farmers.

Role: NLP and Backend Developer. Designed and deployed the retrieval-augmented generation pipeline, conducted prompt engineering, developed the evaluation methodology, and managed cloud infrastructure.

Technologies: Python, Flask, LangChain, Google Gemini API, ChromaDB, RAGAS, Pydantic, Docker, Google Cloud Run, Google Secret Manager, pytest | **Repository:** github.com/walt1210/maize_nlp

YellowMAIze: Mobile-Deployed Maize Disease Diagnosis System *Thesis Project*

Designed and validated a mobile-efficient multi-task deep learning model for maize leaf disease diagnosis, encompassing classification, segmentation, and severity estimation, deployed through TensorFlow Lite for on-device inference.

Role: Helped develop and improve the model pipeline, served as one of two severity raters, and worked on parts of the Android app.

Technologies: PyTorch, TensorFlow Lite, ONNX, scikit-learn, Grad-CAM, Grad-CAM++, Score-CAM, Cohen's Kappa, Spearman Correlation | **Repository:** github.com/LunocoreWired/yellowmaize

Closed-Loop Text-to-SQL Agent *Academic Project*

Developed a producer-critic reflection loop in which a Producer model drafted SQL queries and a Critic model evaluated them against a live database schema, combined with a three-pillar exception handling and recovery cascade for failed queries.

Technologies: Python, Google Gemini API, Pydantic, SQLite | **Repository:** github.com/walt1210/text-to-sql-agent

Autonomous Multi-Agent Deep Research Pipeline *Academic Project*

Built a two-agent CrewAI pipeline pairing a web search Research Planner with a calculator-equipped Synthesis Writer, then used execution tracing to reveal that both agents often bypassed their tools and fabricated figures.

Technologies: Python, CrewAI, Gemini API, DuckDuckGo Search | **Repository:** github.com/walt1210/multi-agent-research-pipeline

EDUCATION

Angeles University Foundation, Bachelor of Science in Computer Science, Major in Data Science (Expected Graduation: 2027)

CERTIFICATIONS

- Reinventing Learning through Artificial Intelligence, Holy Angel University
- Introduction to Cybersecurity, Cisco Networking Academy
- Introduction to Data Science, Cisco Networking Academy
- Microsoft Azure Fundamentals (AZ-900)
- Discovering Entrepreneurship, Cisco Networking Academy
- CCNA: Introduction to Networks, Cisco Networking Academy
- AI Prompt Design, Straits Interactive

Interests: Chess, language learning, photography, music, stargazing

The following presents six projects in greater depth, covering the problem addressed, key features, technologies used, and links to supporting repositories.

MAIze: AI-Powered Maize Disease Diagnosis System (Guidance Module) *Thesis Project*

Farmers need fast, reliable answers to crop disease questions, but general-purpose AI chatbots can generate incorrect or fabricated agricultural advice. This project developed a retrieval-augmented generation module that grounds its recommendations in a curated agricultural knowledge base, deployed as the guidance backend for a mobile diagnostic application designed for Filipino farmers.

- Retrieval pipeline querying a ChromaDB vector store of verified documents before generating a response
- Evaluated with RAGAS and a domain expert review of 30 cases, rated 4.67 out of 5 overall

Technologies: Python, Flask, LangChain, Google Gemini API, ChromaDB, RAGAS, Pydantic, Docker, Google Cloud Run

Link: github.com/walt1210/maize_nlp

YellowMAIze: Mobile-Deployed Maize Disease Diagnosis System *Thesis Project*

Farmers working in the field often lack reliable internet access, making diagnostic tools that depend on cloud inference impractical on site. This project designed and validated a mobile-efficient multi-task deep learning model that performs disease classification, severity estimation, and lesion segmentation from a single leaf photograph, running entirely on-device.

- Three outputs from a single forward pass in about 60 milliseconds on a CPU, via TensorFlow Lite
- Severity estimates compared with independent ratings from two team members (Spearman correlation of 0.875)

Technologies: PyTorch, TensorFlow Lite, ONNX, scikit-learn, Grad-CAM family, Cohen's Kappa

Link: github.com/LunocoreWired/yellowmaize

Closed-Loop Text-to-SQL Agent *Academic Project, Agentic AI*

AI-generated SQL queries can be syntactically valid yet semantically incorrect, and can fail silently when run against a real database schema. This agent addresses that by having a Producer model draft a SQL query and a Critic model evaluate it against the live schema before execution, with a structured recovery process for queries that still fail.

- Producer-critic reflection loop for self-verification before execution
- Three-pillar exception handling and recovery cascade for failed queries

Technologies: Python, Google Gemini API, Pydantic, SQLite

Link: github.com/walt1210/text-to-sql-agent

Autonomous Multi-Agent Deep Research Pipeline *Academic Project, Agentic AI*

A single model asked to both research a topic and synthesize a report often performs both tasks poorly. This pipeline splits the work between two CrewAI agents: a Research Planner equipped with a web search tool, and a Synthesis Writer equipped with a custom calculator tool. Execution tracing revealed that both agents often skipped their tools and stated invented figures with confidence.

- Sequential multi-agent orchestration with full execution tracing across handoffs
- Trace-based analysis exposing tool bypass and fabricated figures behind a polished report

Technologies: Python, CrewAI, Google Gemini API, DuckDuckGo Search

Link: github.com/walt1210/multi-agent-research-pipeline

Subreddit Classification *Academic Project, Supervised Learning*

Conducted multi-class text classification across ten subreddits using a dataset of ten thousand Reddit posts, applying TF-IDF vectorization and Singular Value Decomposition for feature extraction. Compared Logistic Regression, Linear Support Vector Classification, and Random Forest models, with the best-performing model reaching eighty percent accuracy.

Technologies: Python, scikit-learn, pandas, NLTK

Link: *Notebook available on request*

Topic Clustering *Academic Project, Unsupervised Learning*

Performed unsupervised topic discovery on the same Reddit dataset using K-Means and HDBSCAN clustering algorithms, evaluated using Adjusted Rand Index and Silhouette Score, with representative keyword extraction for each identified cluster.

Technologies: Python, scikit-learn, HDBSCAN, pandas, NLTK

Link: *Notebook available on request*

Additional projects, including a relational database design project, are available at github.com/walt1210.